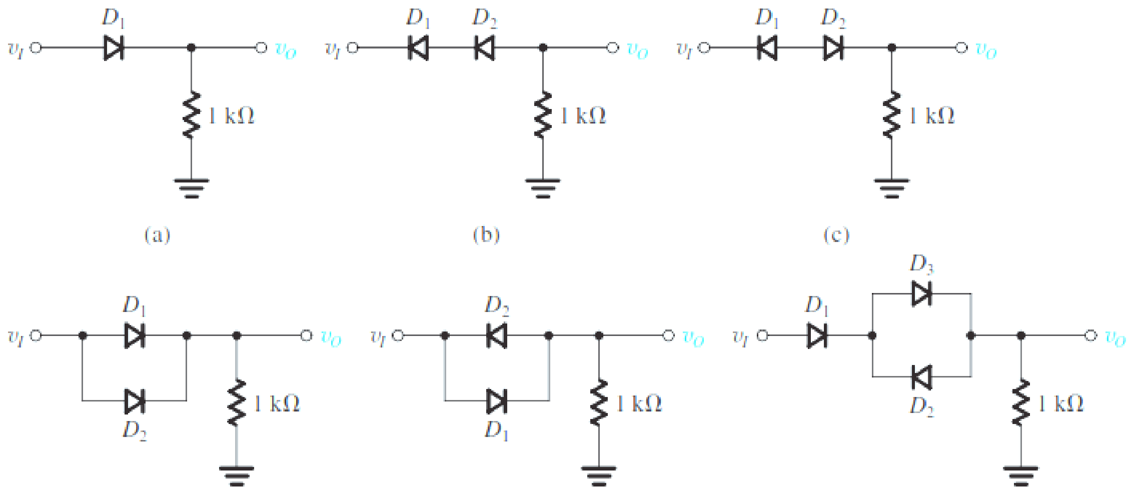
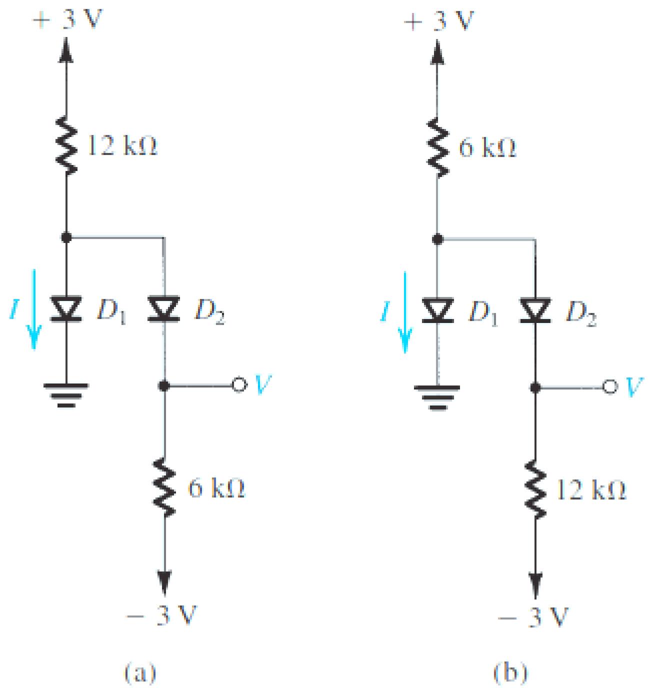
Electronics Engineering
Assignment – 2

1. In each of the ideal-diode circuits shown in Fig. is a 1-kHz, 5-V peak sine wave. Sketch the waveform resulting at V_I. What are its positive and negative peak values?



2. Assuming that the diodes in the circuits of Fig. are ideal, find the values of the labeled voltages and currents.



3. At what forward voltage does a diode conduct a current T equal to 10,000 I_S ? In terms of I_S , what current flows in the same diode when its forward voltage is 0.7 V?
4. Determine I_S for the junction of at T = 400K if A = 100 μm^2 = 20 μm L_n = 20 μm, and L_p = 30 μm. $N_A = 2 \times 10^{16}$ cm^{-3} and $N_D = 4 \times 10^{16}$ cm^{-3}.
5. Determine the junction capacitance of the above device with (a) V_R= 0 V and (b) V_R = 1 V
6. For a p-n junction diode current found to be 2×10^{-17} A for V_R = - 1V, at Hong Kong. If A = 100 μm^2 = 20 μm μ_n = 1350 cm^2 / V - s, and μ_p = 480 cm^2 / V - s. $N_A = 2 \times 10^{16}$ cm^{-3} and N_D =

4×10^{16} cm^{-3}. Determine the temperature of Hong Kong. (Assume that the band gap and thermal voltage remain constant with temperature)

7. For an ideal silicon p-n junction with $N_A = 10^{17}$ cm^{-3} and $N_D = 10^{15}$ cm^{-3}, calculate the built-in V_{bi} at 300 K. Also find the depletion layer width and maximum field at zero bias.

8. Consider a Si p-n junction with n-type doping concentration of 10^{16} cm^{-3} and is forward biased with V = 0.8 V at 300 K. Calculate the minority-carrier hole concentration at the edge of the space charge region on the n-side.

9. Boron is implanted into an n-type Si sample ($N_D = 10^{16}$ cm^{-3}), forming an abrupt junction of square cross section with area = 2 x 10^{-3} cm^2. Assume that the acceptor concentration in the p-type region is $N_A = 4 \times 10^{18}$ cm^{-3}. Calculate V_o, x_n, x_p and E_m for this junction at equilibrium (300 K). Sketch E and the charge density as a function of x.

10. An ideal silicon *p-n* junction has $N_D = 10^{18}$ cm^{-3}, $N_A = 10^{16}$ cm^{-3}, $\tau_p = \tau_n = 10^{-6}$ s, $D_n = 21$ cm^2/s, $D_p = 10$ cm^2/s, and a device area of 1.2 x 10^{-5} cm^2. *(a)* Calculate the theoretical saturation current at 300 K. *(b)* Calculate the forward and reverse currents at ±0.7 V.